Pathologic Interpretation:

A. **Right lower paratracheal lymph node #4R,** [redacted]
- No tumor seen in one lymph node (0/1).

B. **Subcarinal lymph node #7,FS:**
- No tumor seen in one lymph node (0/1).

C. **Lateral bronchial angle lymph node 10,** [redacted]
- No tumor seen in one lymph node (0/1).

D. **Peribronchial lymph node, right upper lobe #11R:**
- One lymph node, no malignancy seen.

E. **Inferior pulmonary ligament lymph node #9:**
- One lymph node, no malignancy seen.

F. **Right lung for bronchial margin,** [redacted]
- Infiltrating poorly differentiated focally keratinizing squamous cell carcinoma, 9.3 cm in greatest dimension invading through pleura into ribs.
- No tumor seen in nine (9) lymph nodes (0/9).
- Pulmonary artery and vein, no tumor seen.
- The tumor is in the bone marrow of the ribs including the superior posterior rib margin.

Tumor Summary:

Specimen Type
- Pneumonectomy

Laterality
- Right

Tumor Site
- Upper lobe
- Middle lobe
- Lower lobe

Tumor Size
- Greatest dimension: 9.3 cm
*Additional dimensions: 8.5 x 8.0 cm

Histologic Type
- Squamous cell carcinoma

Histologic Grade
- G3: Poorly differentiated

Pathologic Staging (pTNM)

Primary Tumor (pT)
- pT3: Tumor of any size that directly invades any of the following: chest wall (including superior sulcus tumors), diaphragm, mediastinal pleura, parietal pericardium; or
Tumor of any size in the main bronchus less than 2 cm distal to the carina but without involvement of the carina; or
Tumor of any size associated atelectasis or obstructive pneumonitis of the entire lung

Regional Lymph Nodes (pN)
- pN0: No regional lymph node metastasis
- Specify: Number examined: 12
  Number involved: 0

Distant Metastasis (pM)
- pMX: Cannot be assessed

Margins
- Margin(s) involved by invasive carcinoma: Ribs **(posterior-superior)**

Direct Extension of Tumor
- Chest wall (including superior sulcus tumors)
- Visceral pleura

Venous (Large Vessel) Invasion (V)
- Absent

Arterial (Large Vessel) Invasion
- Absent

***Lymphatic (Small Vessel) Invasion (L)**
*Present

As the attending pathologist, I attest that I: (i) Examined the relevant preparation(s) for the specimen(s); and (ii) Rendered the diagnosis(es).

Intraoperative Consultation

A. Right lower paratracheal lymph node #4R, No tumor seen in one lymph node (0/1)

B. Subcarinal lymph node #7 : No tumor seen in one lymph node (0/1)

C. Lateral bronchial angle lymph node 10, No tumor seen in one lymph node (0/1)

F. Right Lung, : No carcinoma seen in bronchial margin.

Clinical History:

None given

Operation Performed
Mediastinoscopy, right thoracotomy, right pneumonectomy

Pre Operative Diagnosis:

Lung cancer

Specimen(s) Received:
A: Right lower paratracheal lymph node #4R,

B: Subcarinal lymph node #7,
C: Lateral bronchial angle lymph node 10,
D: Peribronchial lymph node, right upper lobe #11R
E: Inferior pulmonary ligament lymph node #9
F: Right lung for bronchial margin,

Gross Description:

A. Received fresh is a single ovoid soft tissue fragment consistent with a lymph node, 1 x 0.9 x 0.5 cm. The specimen is submitted in toto in one block for frozen section.

B. Received fresh is an irregularly shaped soft tissue fragment, 1 x 0.6 x 0.4 cm. The specimen is bisected and submitted in toto in one block for frozen section.

C. Received fresh are multiple irregularly shaped soft tissue fragments, 2.5 x 2 x 0.5 cm in aggregate. The specimen is submitted in toto in one block for frozen section.

D. Received in formalin is a single ovoid soft tissue fragment, 0.5 x 0.5 x 0.3 cm in greatest dimension. The specimen is bisected and submitted in toto in one block.

E. Received in formalin is an ovoid soft tissue fragment, 1.5 x 0.5 x 0.5 cm in greatest dimension. The specimen is bisected and submitted in toto in one block.

F. Received fresh is a lung resection specimen, which is anatomically distorted, with multiple staple lines. The pleural surface is intact, pale purple-blue without any areas of obvious puckering. There is a main stem bronchus extending from the specimen for a distance of 4 cm. Surrounding this bronchus are multiple subcarinal lymph nodes. Attached to the anterior lateral surface of the specimen are three ribs. The lung measures 21 x 14 x 8 cm in greatest dimension. The attached rib measures 12.5 cm in length and 7.5 cm in aggregate diameter. Upon opening the specimen along the bronchial tree, there is a cystic mass replacing the majority of the lung parenchyma. It measures 9.3 x 8.5 x 8 cm in greatest dimension. The cystic mass is unilocular. The wall of the mass is at most 2.5 cm in thickness. Centrally located within the cyst is hemorrhagic and necrotic fluid. Upon removing the fluid, a large papillary necrotic mass is identified measuring at most 5 cm in greatest dimension. This lesion grossly appears to approach the pleura at the right superior lateral aspect close to the ribs and is grossly at a distance of 0.1 cm from the inked pleural surface. Grossly, this lesion appears to focally involve the surrounding portion of rib in the central portion of the length of the rib and is at least 2.5 cm from the anterior bone resection margin and at least 2 cm from the lateral posterior bone resection margin. The pulmonary parenchyma surrounding the cavitary lesion is hemorrhagic and consolidated. No other lesions are grossly identified. Sections are submitted as follows:

1	Bronchial resection margin for frozen section
2&3	Sections of cavitary lesion in relation to closest inked parietal pleura
4&5	Composite section through greatest thickness of cavitary wall
6	Section of lesion in relation to large vessels
7&8	Additional sections of lesion
9	Section of lesion in relation to surrounding congested and consolidated pulmonary parenchyma
10-13	One possible lymph node per block
14	Two possible lymph nodes
15-18	One possible lymph node in toto
19	Pulmonary artery and vein
20-23	Section of lesion in relation to bone (for decalcification)
24	Rib margin, anterior superior
25	Rib margin, anterior middle
26	Rib margin, anterior inferior
27	Rib margin, posterior superior
28	Rib margin, posterior middle
29	Rib margin, posterior inferior.